INTERNSHIP PRESENTATION

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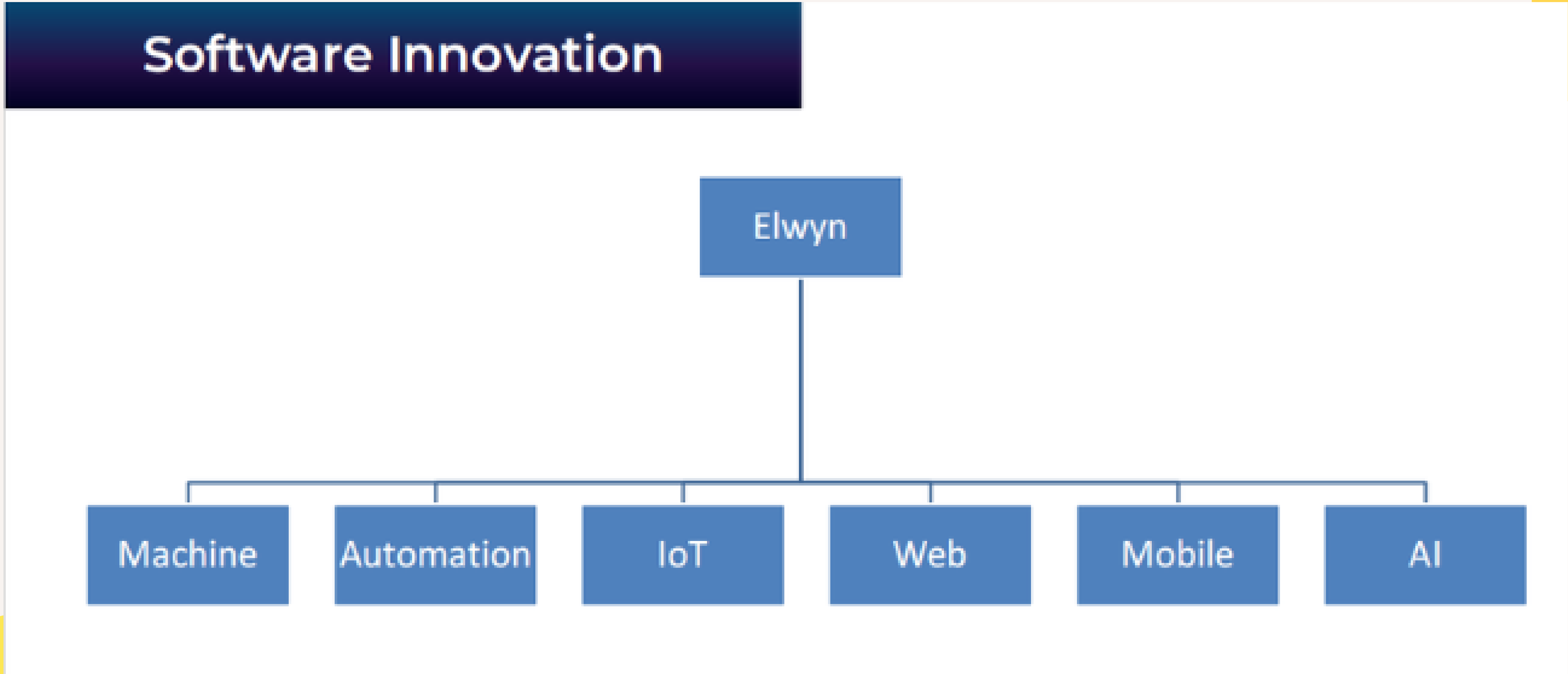
Faculty: FICT

COMPANY BACKGROUND

- Sophic Automation Sdn Bhd was founded in 2007 with the mission of providing the most innovative and effective Industrial Automation Solutions to customers.

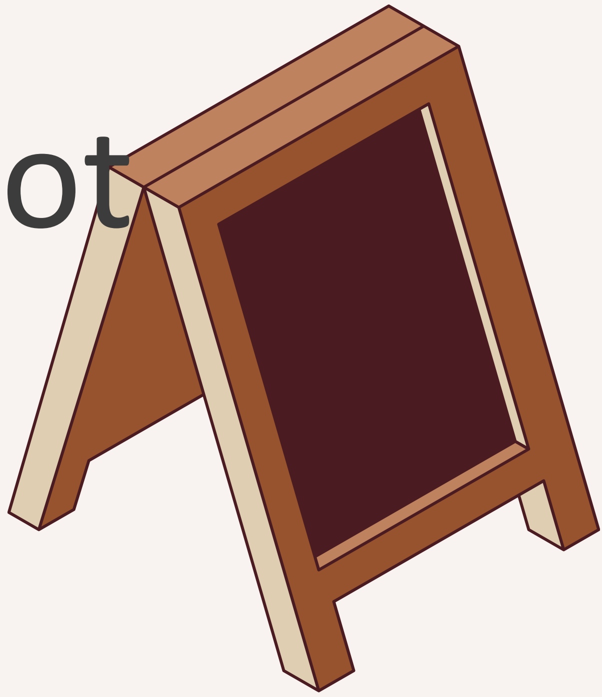
- Sophic offer an attractively priced and complete outsourcing mechanism, through which sophic develop, supply, and manage customized automation projects for customers.

JOB FUNCTION



JOB FUNCTION

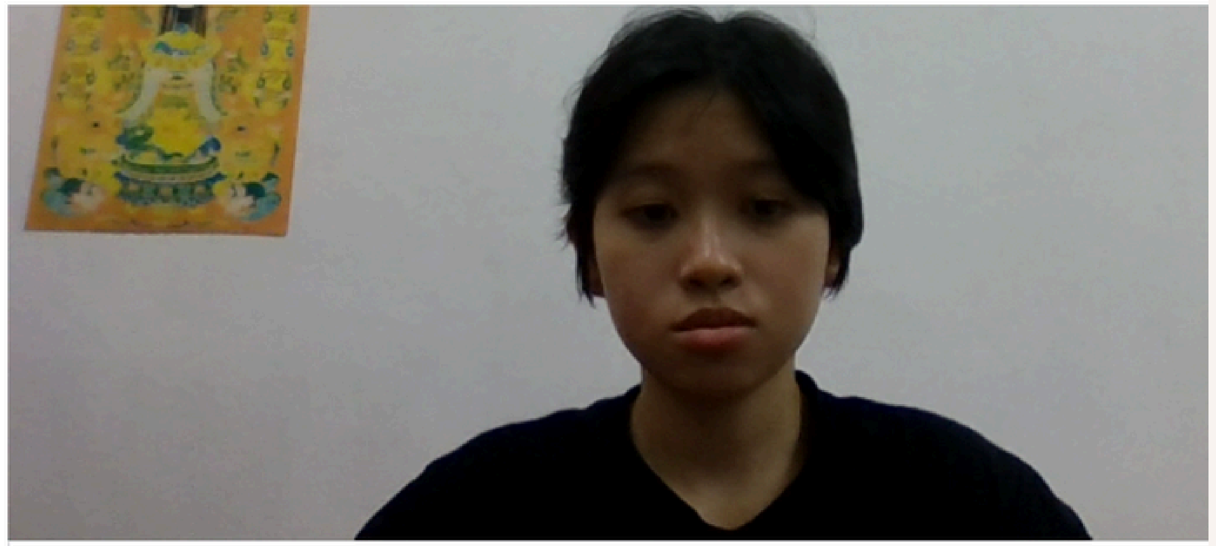
- **Data Extraction:** Extract data from multiple sources such as textual, web pages, API, formatted report.
- **Data Conversion:** Standardize formats (CSV → JSON) for processing and storage.
- **REST API:** Build Flask-based APIs for real-time queries and results delivery.
- **Data Analysis:** Analyze fluctuations, identify root causes, and develop actionable insights.



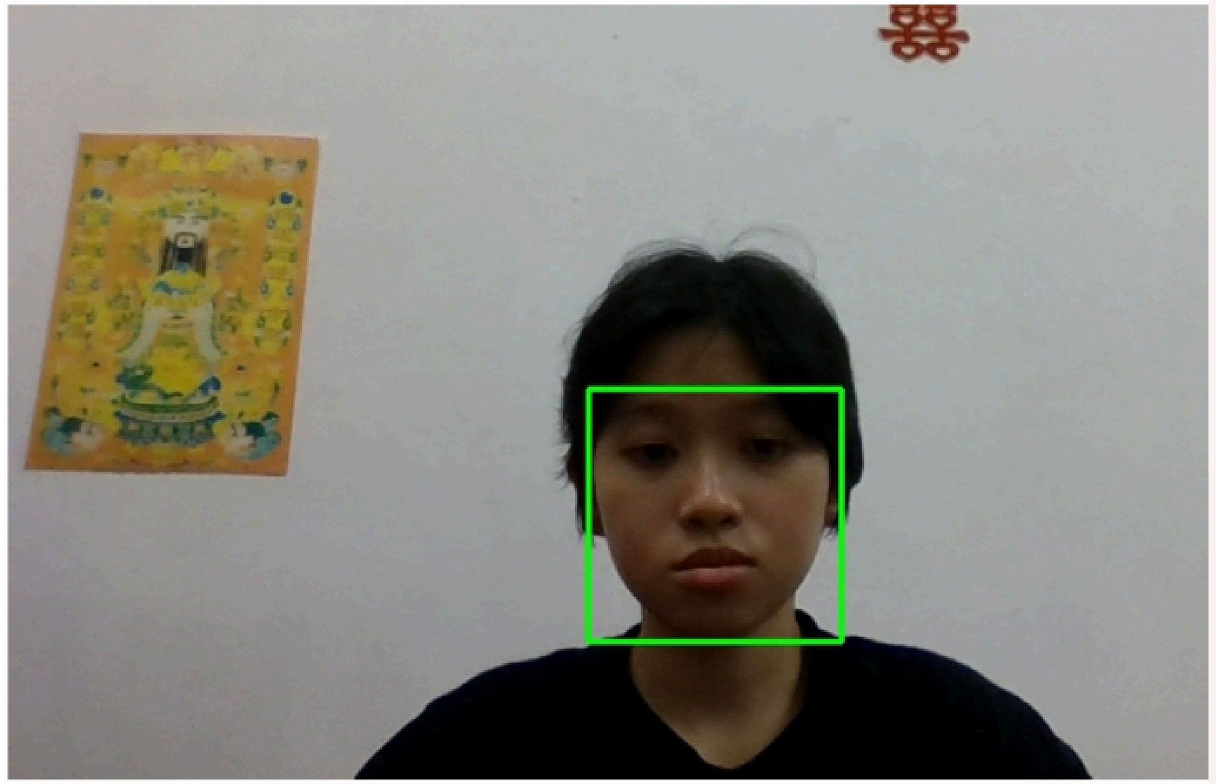
WHAT HAS BEEN DONE BY INTERN DURING THE INDUSTRIAL TRAINING

STREAMLIT

Objectives: Create user profiles by recognizing faces



✕ Clear photo



Captured Image with Face Detection

Welcome, yueming! Your profile has been created successfully!

User Information

Name: yueming

Age: 33

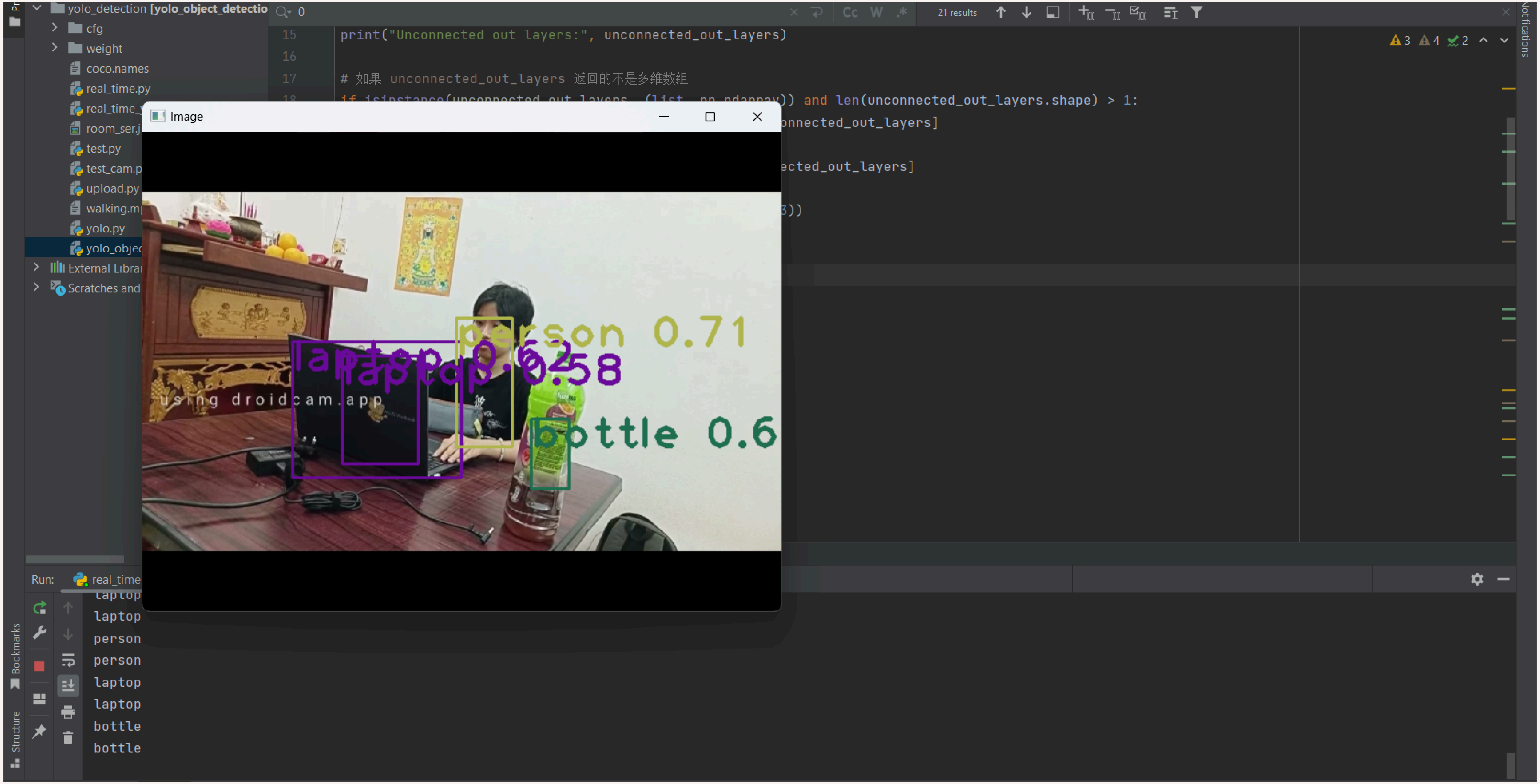
Email: asdf@gmail.com

Registered: 2024-11-06

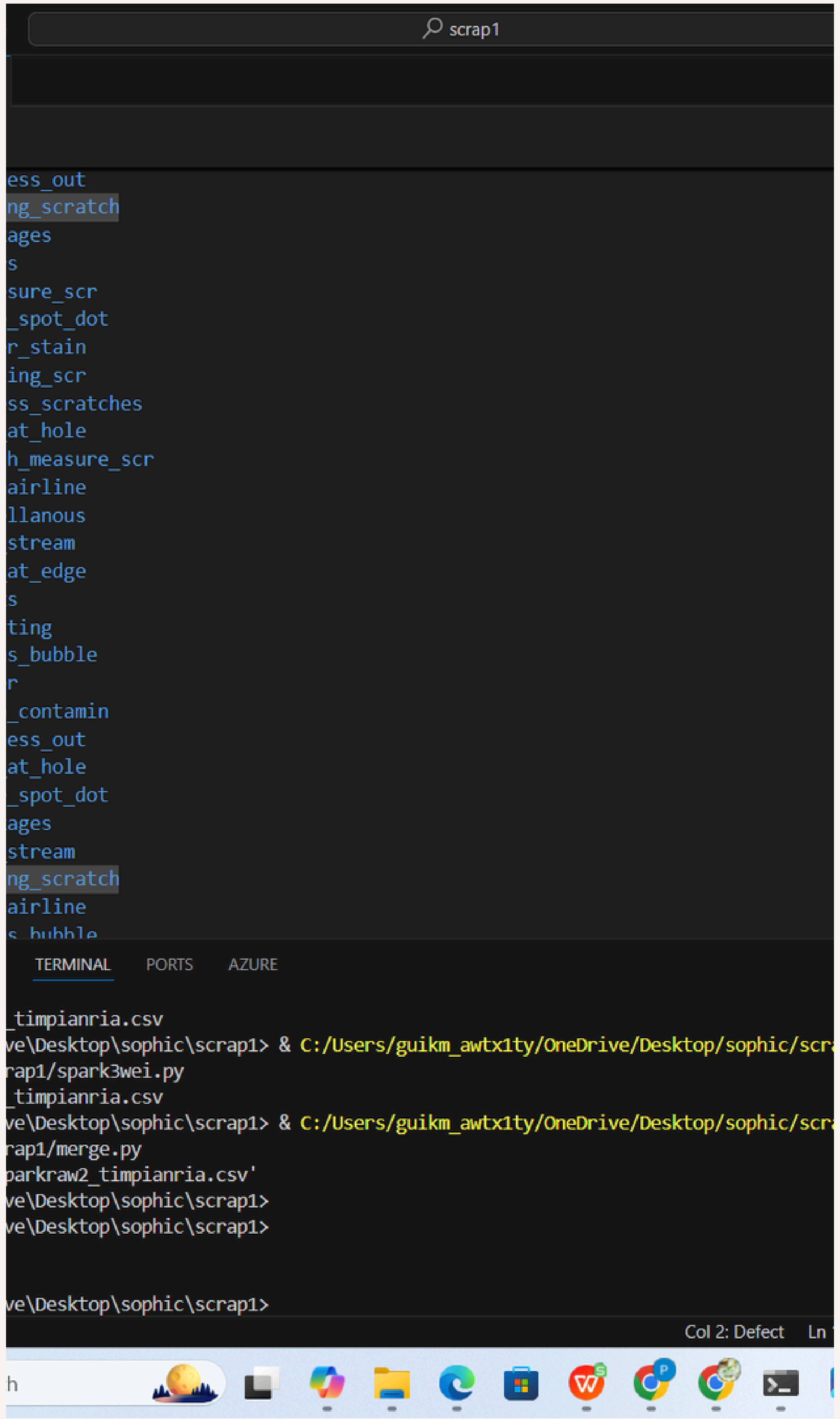
Profile Photo - yueming

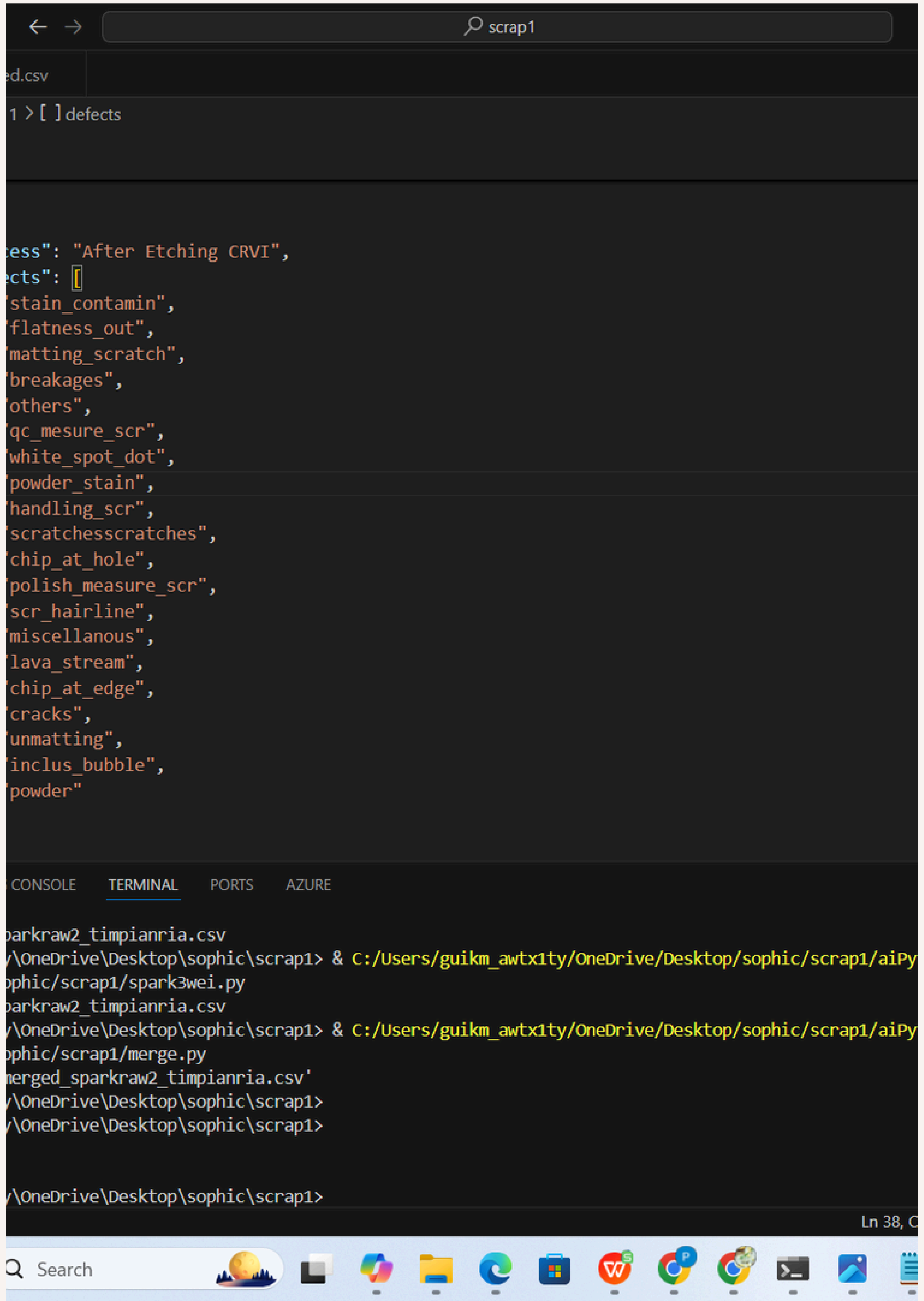
Log Out

OBJECT-DETECTION SYSTEM

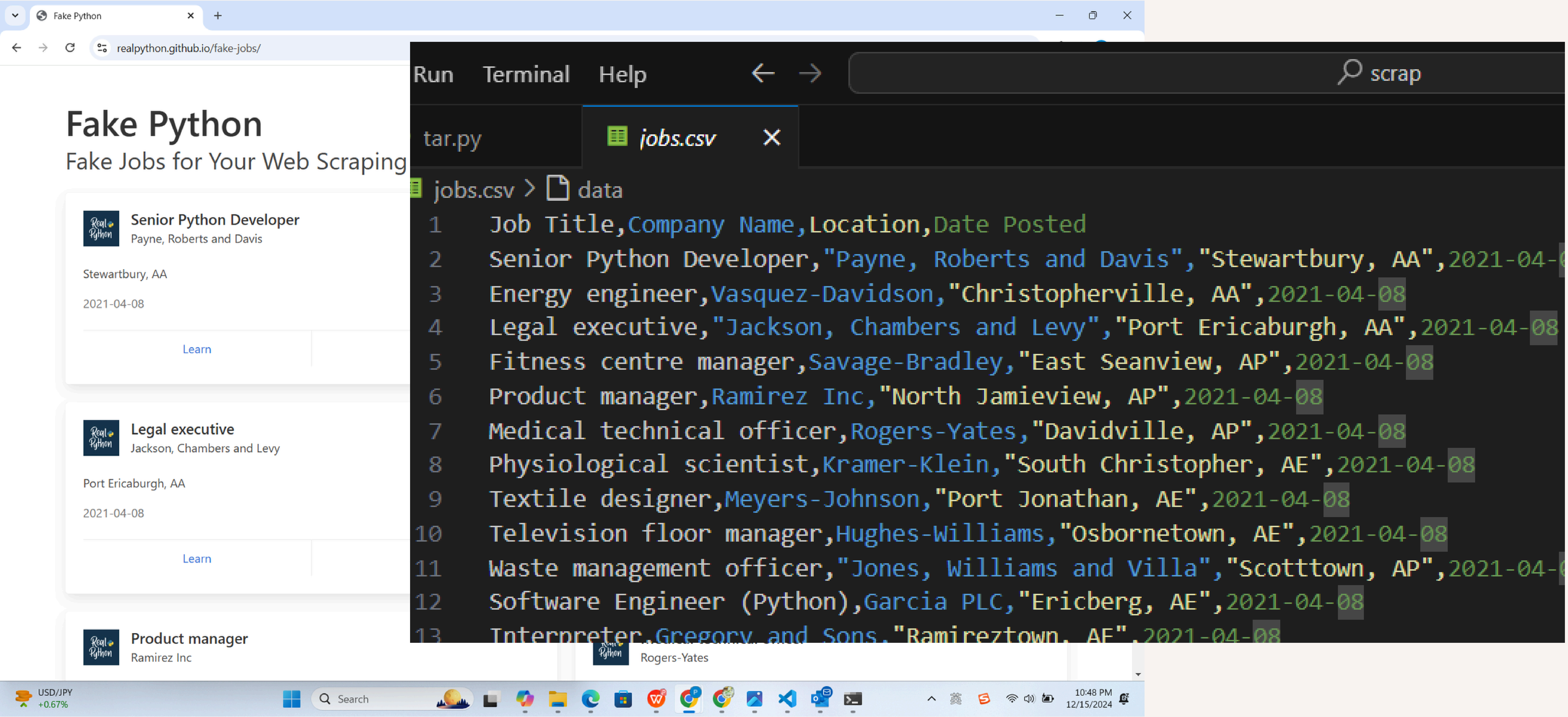


TRANSFER CSV TO JSON

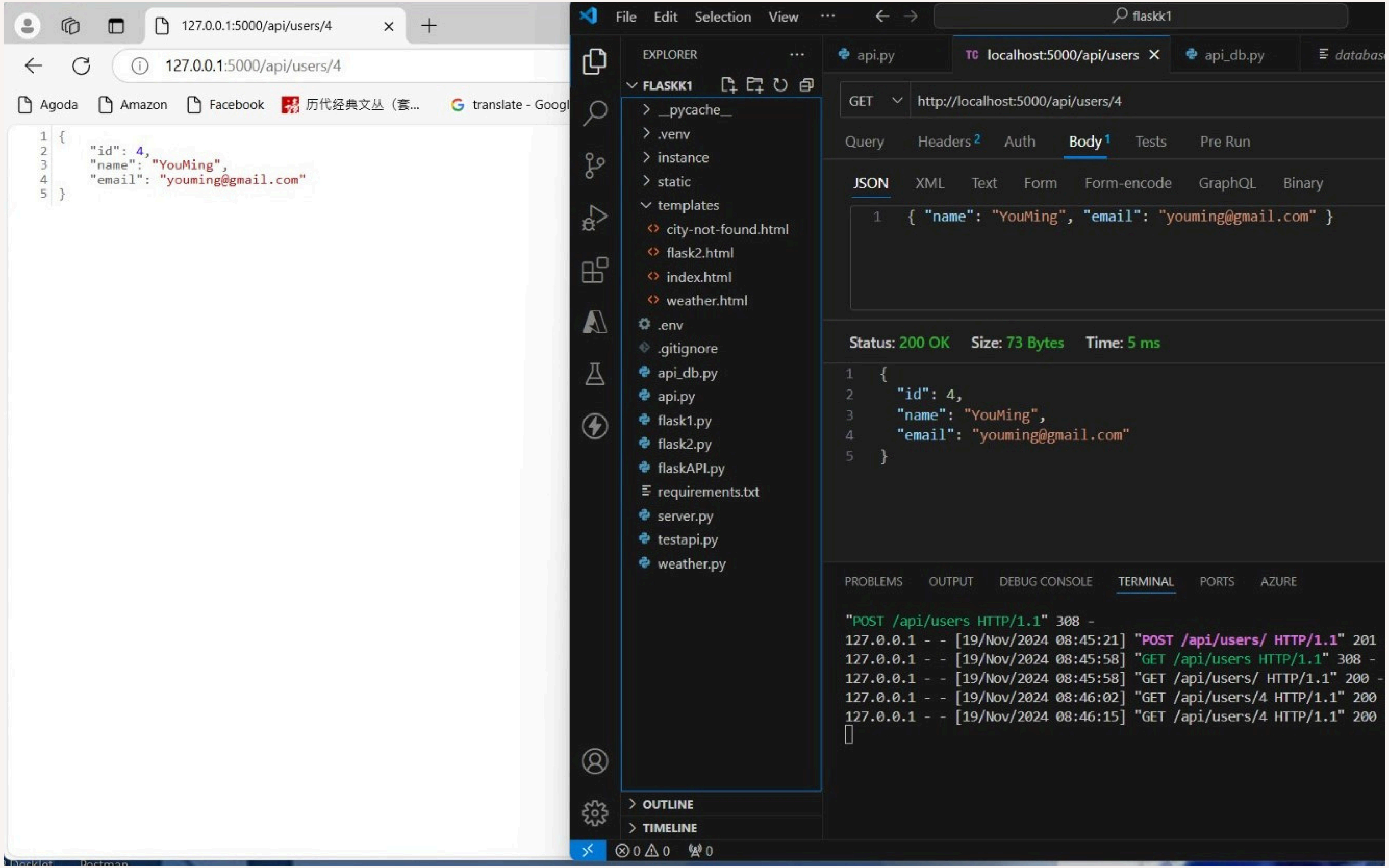




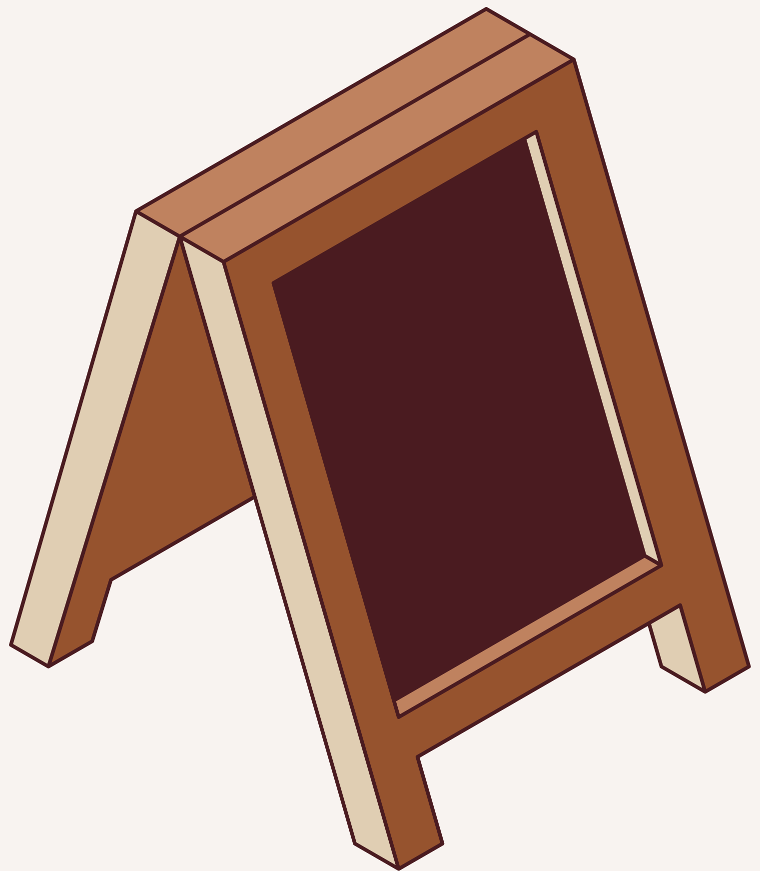
WEB SCRAPING



FLASK REST API



RELATIONSHIP BETWEEN WATER LEVEL OF PUMPHOUSE AND WEATHER



RAW DATA

id	battery_vol	latest_data	latest_update
1	3.6	0.413	3/23/2023 17:22
2	3.6	0.413	3/23/2023 17:24
3	3.6	0.413	3/23/2023 17:26
4	3.6	0.413	3/23/2023 17:28
5	3.6	0.413	3/23/2023 17:31
6	3.6	0.413	3/23/2023 17:33
7	3.6	0.413	3/23/2023 17:35
8	3.6	0.413	3/23/2023 17:37
9	3.6	0.412	3/23/2023 17:39
10	3.6	0.412	3/23/2023 17:41
11	3.6	0.413	3/23/2023 17:43
12	3.6	0.412	3/23/2023 17:45
13	3.6	0.412	3/23/2023 17:47
14	3.6	0.412	3/23/2023 17:50
15	3.6	0.412	3/23/2023 17:52
16	3.6	0.412	3/23/2023 17:54
17	3.6	0.412	3/23/2023 17:56
18	3.6	0.412	3/23/2023 17:58
19	3.6	0.412	3/23/2023 18:00
20	3.6	0.412	3/23/2023 18:02
21	3.6	0.412	3/23/2023 18:05
22	3.6	0.412	3/23/2023 18:07
23	3.6	0.412	3/23/2023 18:09
24	3.6	0.412	3/23/2023 18:11
25	3.6	0.412	3/23/2023 18:13
26	3.6	0.412	3/23/2023 18:15
27	3.6	0.412	3/23/2023 18:17
28	3.6	0.412	3/23/2023 18:19
29	3.6	0.412	3/23/2023 18:22
30	3.6	0.412	3/23/2023 18:24

MERGE DATA WITH WEATHER

battery_vol	latest_data	latest_upda	temp	precip	sealevelpre	solarradiati	solarenergy	severerisk	icon	description
3.6	0.413	3/23/2023	83.2	0.034	1008.1	292	25.3	60	rain	Partly cloudy throughout the day with rain in the morning and afternoo
3.6	0.408	3/24/2023	82.9	0.017	1008.4	299.3	25.8	60	rain	Partly cloudy throughout the day with afternoon rain.
3.6	0.402	3/25/2023	83.1	0	1009.1	301.5	25.9	60	partly-clouc	Partly cloudy throughout the day.
3.6	0.402	3/26/2023	83.6	0.003	1008.3	300.1	25.8	60	rain	Partly cloudy throughout the day with morning rain.
3.6	0.402	3/27/2023	82.6	0.175	1008.1	243.7	21.1	60	rain	Partly cloudy throughout the day with rain.
3.6	0.406	3/28/2023	82.3	0	1009.1	295.2	25.5	60	partly-clouc	Partly cloudy throughout the day.
3.6	0.407	3/29/2023	83.2	0	1009.5	303.2	26.2	60	partly-clouc	Partly cloudy throughout the day.
3.6	0.414	3/30/2023	83.8	0.01	1008.5	313.6	27	30	rain	Partly cloudy throughout the day with morning rain.

visualcrossing

Weather Data | Weather API | Query Builder | Pricing | API Docs | More

Search docs...

Sign in | Sign up

Addresses, partial addresses or lat,lon | History or forecast data | Data units

Query options

Data sections | Weather elements | Degree days | Wind & solar | Agriculture | Weather stations

API | Grid | Chart | JSON | CSV

Weather Data: Daily | Hourly

Additional Data: Current | Events | Alerts

Data Details: Info | Stations

Questions about the data? | Download

Available weather data for **Butterworth, Pulau Pinang, Malaysia**. These results are filtered by your query options.

datetime	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	humidity	precip	precipprob	precipcover	preciptype	snow	snowdepth	windgust	windspeed	winddir	pr
2024-12-15	89.4	77	82.4	98.1	77	87.6	74.4	77.4	0.084	100	37.5	rain	0	0	6.9	7.8	343.6	10
2024-12-16	85.7	74	78.7	96	74	81.8	74.8	88.3	0.263	96.8	54.17	rain	0	0	6.9	4.9	38.1	10
2024-12-17	80	73.7	76.2	80	73.7	76.2	73.6	91.8	0.26	90.3	41.67	rain	0	0	8.7	5.1	51	10
2024-12-18	80.1	72.1	75.3	84.9	72.1	75.7	72.4	90.8	0.253	100	54.17	rain	0	0	8.5	5.1	98.2	10
2024-12-19	75.6	71.7	73.6	75.6	71.7	73.6	71.7	93.9	0.45	87.1	58.33	rain	0	0	12.1	4.5	61.6	10
2024-12-20	80.5	72.4	76	85.5	72.4	76.4	72.3	88.6	0.166	93.5	20.83	rain	0	0	3.6	3.8	89.9	10
2024-12-21	86.3	72.4	77.6	94.2	72.4	79.9	72.8	85.9	0.217	96.8	20.83	rain	0	0	5.4	4	80.8	10
2024-12-22	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2024-12-23	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2024-12-24	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

1 of 2 pages. Sign in to see the full dataset.

[8]:
```
df.describe()
```

[8]:

	battery_voltage	latest_data	temp	precip	sealevelpressure	solarradiation	solarenergy	severerisk
count	5.020000e+02	502.000000	502.000000	502.000000	502.000000	502.000000	502.000000	502.000000
mean	3.600000e+00	0.236145	83.649402	0.083217	1009.355976	234.531275	20.251793	40.866534
std	3.022819e-14	0.636087	2.009704	0.199908	1.406612	64.045603	5.529780	18.147657
min	3.600000e+00	0.000000	77.900000	0.000000	1005.500000	23.900000	2.200000	10.000000
25%	3.600000e+00	0.019000	82.200000	0.000000	1008.400000	211.525000	18.200000	30.000000
50%	3.600000e+00	0.021000	83.600000	0.000000	1009.400000	254.550000	22.000000	30.000000
75%	3.600000e+00	0.073000	85.000000	0.038000	1010.200000	279.800000	24.200000	60.000000
max	3.600000e+00	9.999000	88.300000	1.472000	1013.700000	313.600000	27.100000	100.000000

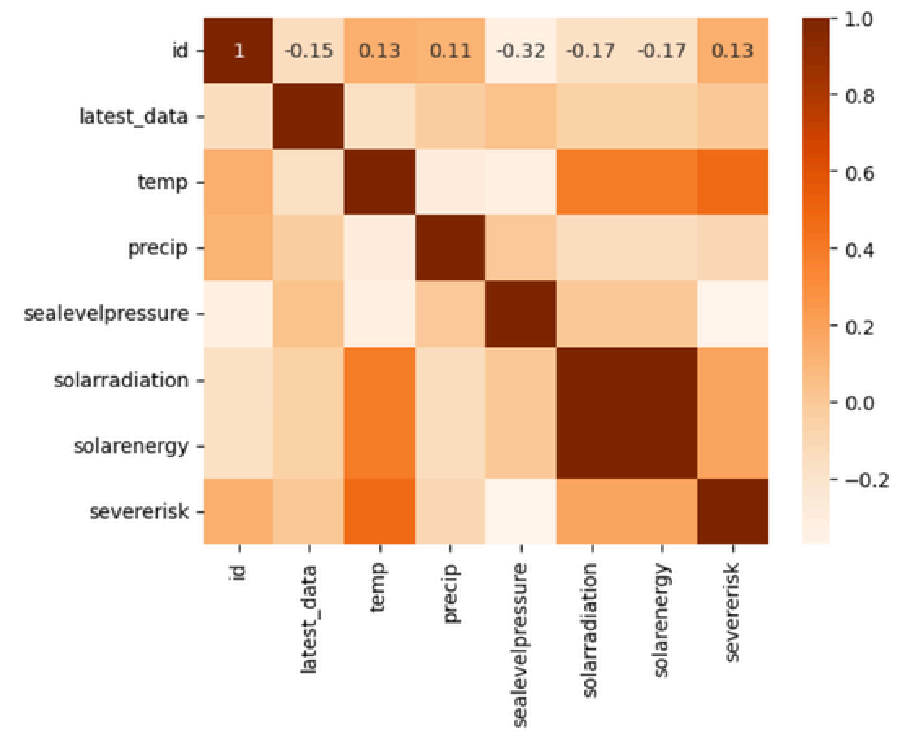
[14]:
```
df.describe().T
```

[14]:

	count	mean	std	min	25%	50%	75%	max
battery_voltage	502.0	3.600000	3.022819e-14	3.6	3.600	3.600	3.600	3.600
latest_data	502.0	0.236145	6.360868e-01	0.0	0.019	0.021	0.073	9.999
temp	502.0	83.649402	2.009704e+00	77.9	82.200	83.600	85.000	88.300
precip	502.0	0.083217	1.999084e-01	0.0	0.000	0.000	0.038	1.472
sealevelpressure	502.0	1009.355976	1.406612e+00	1005.5	1008.400	1009.400	1010.200	1013.700

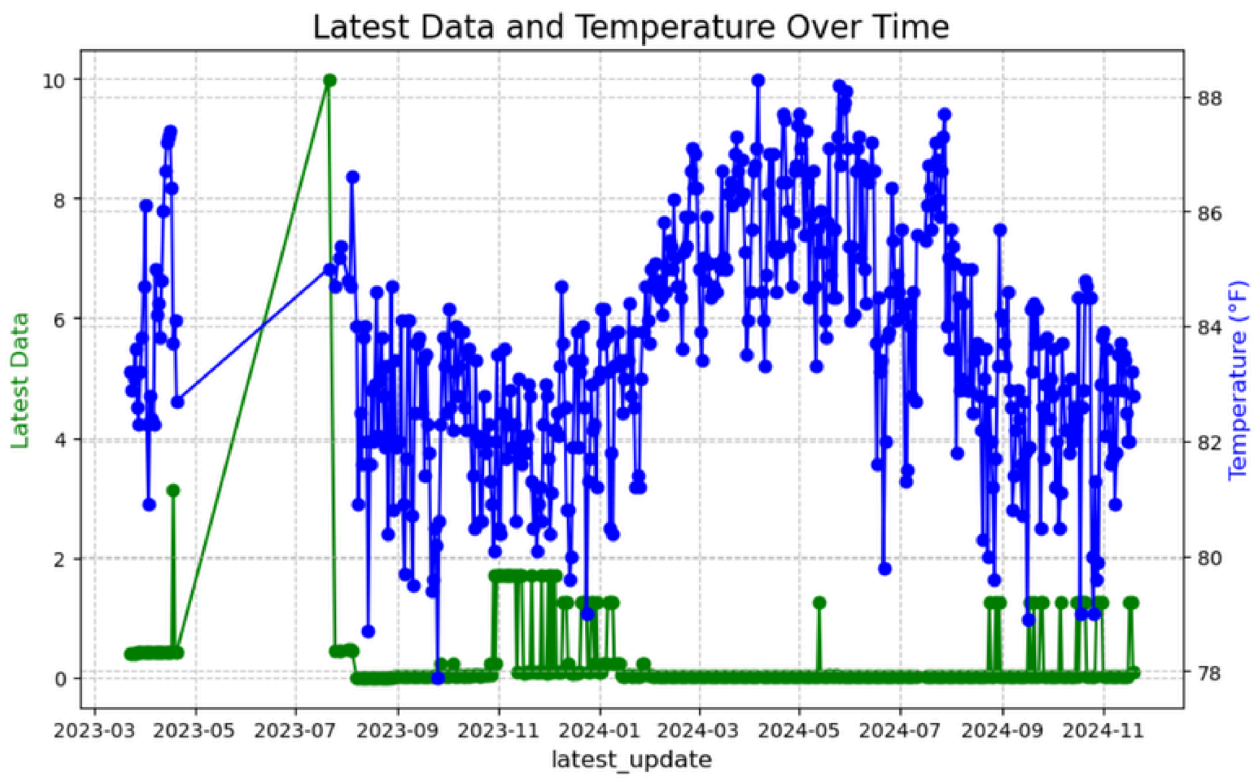
[40]:
```
#create a heatmap for visualizing the correlation matrix
sns.heatmap(corr_matrix, annot=True, cmap="Oranges")
```

[40]: <Axes: >



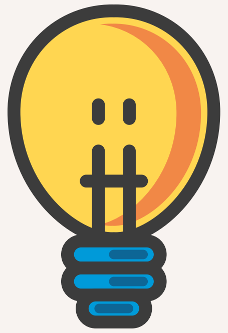
[ ]:

[42]:
```
print(df['latest_data'].unique())
```



[ ]:

stationery of time series, how to check. determine the value of d, p, q



ARIMA TIME SERIES MODEL

Accuracy Metrics for Time Series Forecast

```
# Define forecast accuracy function
def forecast_accuracy(forecast, actual):
    forecast = np.array(forecast)
    actual = np.array(actual)
    mape = np.mean(np.abs(forecast - actual) / np.abs(actual))  # MAPE
    me = np.mean(forecast - actual)                             # ME
    mae = np.mean(np.abs(forecast - actual))                    # MAE
    mpe = np.mean((forecast - actual) / actual)                 # MPE
    rmse = np.mean((forecast - actual) ** 2) ** .5              # RMSE
    corr = np.corrcoef(forecast, actual)[0, 1]                  # corr
    mins = np.amin(np.hstack([forecast[:, None], actual[:, None]]), axis=1)
    maxs = np.amax(np.hstack([forecast[:, None], actual[:, None]]), axis=1)
    minmax = 1 - np.mean(mins / maxs)                           # minmax
    acf1 = acf(forecast - actual)[1]                            # ACF1
    return {'mape': mape, 'me': me, 'mae': mae, 'mpe': mpe, 'rmse': rmse, 'acf1': acf1, 'corr': corr, 'minmax': minmax}

# Calculate forecast accuracy
arima_accuracy = forecast_accuracy(fc_series, test_single)
print(f'ARIMA accuracy: {arima_accuracy}')
```

```
ARIMA accuracy: {'mape': 1.9150531166010705, 'me': -0.30710163575657917, 'mae': 0.3117743198056307, 'mpe': 1.5884273183596416, 'rmse': 0.812435041984011
7, 'acf1': 0.5266005260548933, 'corr': 0.019081415243055407, 'minmax': -1.6021873948813394}
```

WHAT HAS THE INTERN LEARNT IN THE INDUSTRIAL TRAINING

(THINGS THAT INTERN HAS NOT LEARNT IN UTAR)

SOFT SKILLS

- **Proactive Attitude**: Learned the importance of asking questions when encountering challenges or uncertainties.
- **Independent Research Skills**: Developed the ability to effectively utilize online resources to solve problems and gain new knowledge.
- **Time Management**: Improved the skill of organizing and prioritizing tasks to meet deadlines efficiently.

HARD SKILLS

- Object Detection
- Web Scraping
- Flask REST API
- Water Level Analysis

WHAT INTERN HAS APPLIED IN THE INDUSTRIAL TRAINING

(THINGS THAT INTERN HAS LEARNT FROM UTAR)

- Jupyter (Python)
- Visual Studio Code (Python)

Thank You